

Anish Patnaik

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EDUCATION

University of California San Diego

Master of Science in Data Science, GPA: 4.0/4.0

Coursework: Web Mining and Recommender Systems, Scalable Data Systems, ML Systems, Efficient AI

Sep 2025 - Dec 2026

San Diego, CA

Manipal Institute of Technology

Bachelor of Engineering in Electronics and Communication, GPA: 9.39/10

Minor Specialization: Data Science

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Data Science

Jul 2018 - May 2022

Manipal, India

SKILLS

Languages: Python, Java, TypeScript, C++, SQL, R

Cloud Development Tools: AWS, Lambda, SQS, SNS, S3, DynamoDB, CloudWatch, IAM, Azure Functions, Azure Blob Storage, Azure Synapse, Azure SQL, Azure ML Studio

Backend and Systems: REST APIs, Microservices, Serverless Architecture, Distributed Systems, System Design, Databases

Tools and DevOps: Docker, Kubernetes, Git, GitHub, CI/CD, GitHub Actions, Jenkins, Terraform, Agile

Machine Learning: PyTorch, TensorFlow, Transformers, RAG, LLM Integration, Scikit-learn, Deep Learning

PROFESSIONAL EXPERIENCE

Amazon

Software Development Engineer Intern

Jun 2026 - Sep 2026

Seattle, WA

- Designing dedicated sync infrastructure with a direct **Lambda invocation** path for user-triggered ownership updates across **3M+ AWS accounts daily**, letting urgent requests bypass **SQS** batch backlogs and return feedback in **seconds**.
- Implementing a **DynamoDB** freshness table with **conditional writes** so older S3 snapshot and event-driven updates cannot overwrite ownership data that was just refreshed from a live source after an on-demand sync.
- Refactoring scheduled, event-driven, and on-demand sync flows to carry source timestamps and freshness metadata through each sync path, keeping ownership updates ordered across live API reads, daily snapshots, delayed notifications, and retries.
- Designing **integration tests** to validate end-to-end sync behavior across live reads, snapshot reads, retries, DLQ handling, and stale-write prevention before rollout.
- Monitoring service traffic and queue-depth patterns with CloudWatch dashboards and alarms for **p50/p99 latency**, Lambda errors, throttles, and reserved concurrency planning, plus rollout safeguards for canary deployment and feature-flag rollback.

Optum - UnitedHealth Group

Associate Data Scientist

Jul 2022 - Aug 2025

Hyderabad, India

- Built an end-to-end extraction pipeline with OpenAI APIs and DiT models to automate appeal processing, reduce manual review work by **15%**, and save approximately **\$1.4M annually**.
- Architected a serverless Azure Functions microservice for access risk reviews, serving **100k+ requests** with **under 50 ms latency** through optimized internal API calls, structured logging, and latency tracking.
- Developed a **96.1% AUC** access recommendation service using peer access similarity features for **400k+ employees**, exposing model outputs through internal APIs and review tools.
- Built reusable REST API and database integration patterns for access risk scoring, letting internal review systems retrieve model-backed recommendations through consistent APIs.
- Reduced new hire attrition from **18% to 11%**, saving approximately **\$0.5M annually**, by implementing EDA workflows across hiring data from **29 business lines** and surfacing actionable retention drivers.
- Created SQL-backed reports and Power BI dashboards that helped business teams review model findings, access risk trends, retention planning, and quarterly performance summaries.

PROJECTS

MyTorch: Custom Deep Learning Library

Dec 2025 - Jan 2026

- Built an object-oriented Python deep learning framework with computational graphs, automatic differentiation, backpropagation, and a custom SGD optimizer for first-principles neural network training without relying on high-level autograd internals.
- Designed modular Linear, Batch Normalization, and Activation layers for training multilayer perceptrons with reusable components for ablation-style experiments.
- Implemented reusable training loops and tensor abstractions to validate model behavior against PyTorch-style neural network mechanics across forward and backward passes with gradient checks.
- Separated layers, optimizers, and activation functions into testable modules, making model definitions easier to extend as new layer types and experiments were added.

UCSD SMASH x NSF HDR ML Challenge

Jan 2026

- Secured **2nd place** by building a reproducible inference pipeline for high-dimensional environmental sensor data, improving robustness on unseen coastal regions during Codabench leaderboard evaluation.
- Optimized preprocessing and evaluation logic to handle geographic distribution shifts, noisy sensor measurements, and hidden test splits.
- Standardized feature processing so local experiments, validation runs, and leaderboard submissions used the same data path from notebooks through final submission.

Neonatal Seizure Detection from EEG Signals

Jan 2024 - Nov 2024

- Built a real-time seizure detection pipeline for **12,000+ EEG samples**, achieving **97.49% accuracy** with a CNN-Attention ensemble and ICA-based preprocessing for noisy neonatal EEG signals.
- Engineered chunking, windowing, and normalization workflows to convert noisy physiological EEG data into stable model-ready features for seizure detection.